



OMNİKON

NATIONAL HACKATHON 2026

TEAM NAME

AlphaX duo

MEMBERS

Mounika A , Shanmugapriya S B

THEME

BioTech & HealthTech

PROBLEM STATEMENT

Conventional prosthetics are costly and often lack adaptive grip control, limiting daily functionality for underprivileged amputees. Although affordable and open-source designs exist, they are scattered and hard to discover. Amputees also struggle to find reliable local makers, get proper fitting guidance, and receive aftercare support. As a result, many functional low-cost limbs are either never accessed or abandoned after a short period.

PROBLEM IDEA

PROPOSED SOLUTION

ProstheticTrust is a mobile-first software platform that connects underprivileged amputees with:

- Curated affordable & open-source prosthetic designs
- Verified local makers / 3D-printing workshops
- Step-by-step fitting and training guidance

Focus (MVP) : Upper-limb (below-elbow) prosthetics

- Smart Design Catalogue with Mechanical quality scoring
- Guided residual limb assessment (photos + simple measurements)
- Verified local maker network with ratings and proof
- Visual + voice fitting & training modules
- Trust system (verification, staged process, feedback)

How It Addresses The Problem

- Solves discovery problem → users find suitable low-cost designs easily
- Solves fitting problem → clear guided process reduces pain and abandonment
- Solves trust problem → verified makers + quality signals
- Solves aftercare problem → continuous support and repair guidance

Innovation And Uniqueness

- Mechanical Quality Evaluation of every design (strength, grip usefulness, durability, suitability for Indian activities)
- Trust-first system designed specifically for low-cost market fears
- Fitting and training treated as a core feature, not an afterthought
- Strong emphasis on reducing prosthetic abandonment
- Built with deep understanding of both software and mechanical realities

SYSTEM ARCHITECTURE

Technologies

- Frontend: Flutter (Android-first)
- Backend: Firebase (or Node.js + MongoDB)
- Storage: Cloud Storage + Local caching (Offline support)
- Languages: Tamil + English
- Other: Image upload, measurement forms, voice guidance

Methodology & Process

- User Registration & Need Selection
- Residual Limb Photo + Measurement
- Design Recommendation (filtered by Mechanical scores)
- Local Maker Matching (Verified makers)
- Order & Staged Process
- Fitting Guidance Modules
- Feedback & Rating



ProsthetiTrust

Affordable Prosthetic Platform



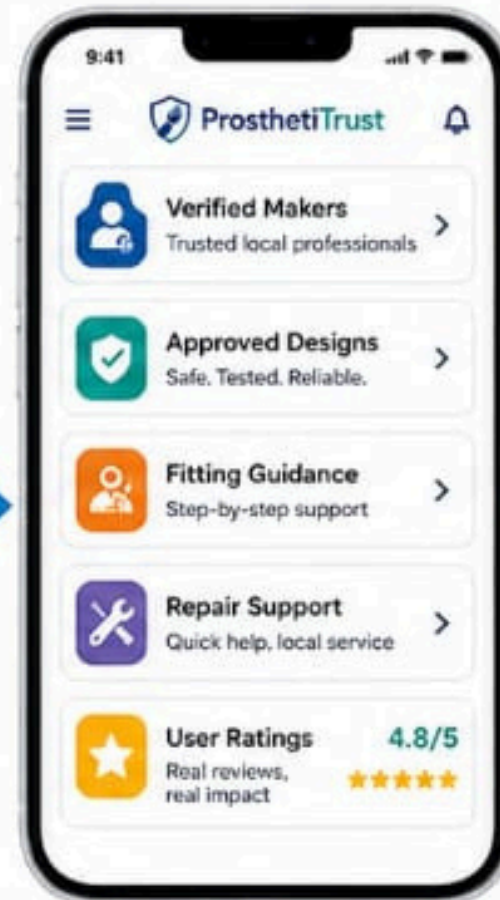
USER

Amputee / Caregiver



User Needs

- Affordable
- Reliable
- Comfortable
- Local Support



Verified Maker

- Background Verified
- Quality Assured
- Trained Professionals



VERIFIED LOCAL MAKER
Skilled • Verified • Trusted



Safe Fitting

- Custom Fit
- Comfort Check
- Guidance & Training



PROSTHETIST
Assess • Fit • Train



3D PRINTER
Precision • Quality • Local



PROSTHETIC
Functional • Durable
Empowering



Delivery

- On-time Delivery
- Secure Packaging
- Ready to Use



Follow-up

- Usage Check
- Adjustments
- Ongoing Support

BUILT ON TRUST. BACKED BY SYSTEM.



**TRUST
LAYER**



QUALITY CHECK

Every device passes
multi-point inspection



STAGED PAYMENT

Pay in milestones.
Safe & transparent.



WARRANTY

Standard warranty on
every device.



SPARE PARTS

Easy access to
genuine spare parts.



15/45/90-DAY FEEDBACK

Scheduled check-ins
to ensure satisfaction.



**SCORE
8.7/10**
★★★★★

MECHANICAL QUALITY SCORE

Data-driven score
you can trust.



Affordable
Solutions



Local Makers,
Local Impact



Empowering Lives.
Restoring Independence.



Technology + Trust
for a Better Tomorrow.

FEASIBILITY AND VIABILITY

Feasibility Analysis

- Pure software solution — no hardware manufacturing required
- Open-source designs already available
- Local 3D-printing ecosystem exists in many towns
- MVP can be built by focusing only on upper-limb designs
- Manual verification of makers in the first phase

Potential Challenges & Risks

- Building initial trust with users and makers
- Ensuring consistent quality from local makers
- Low digital literacy in target users
- Internet connectivity issues in rural areas

Strategies To Overcome

- Start with limited verified makers and curated designs
- Strong visual + voice interface
- Offline-first core features
- Staged process guidance and rating system for trust
- Keep the interface very simple

IMPACT AND BENEFITS

Impact And Benefits

Target Audience

- Underprivileged upper-limb amputees (especially rural)
- Local makers and 3D-printing workshops
- NGOs and community health workers

Benefits

- Social: Restores independence, dignity, and ability to work
- Economic: Lower cost access to functional limbs + income opportunity for local makers
- Systemic: Reduces abandonment of prosthetics and improves utilization of existing low-cost designs
- Scalable: Can expand to more designs, languages, and regions

RESEARCH AND REFERENCES

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